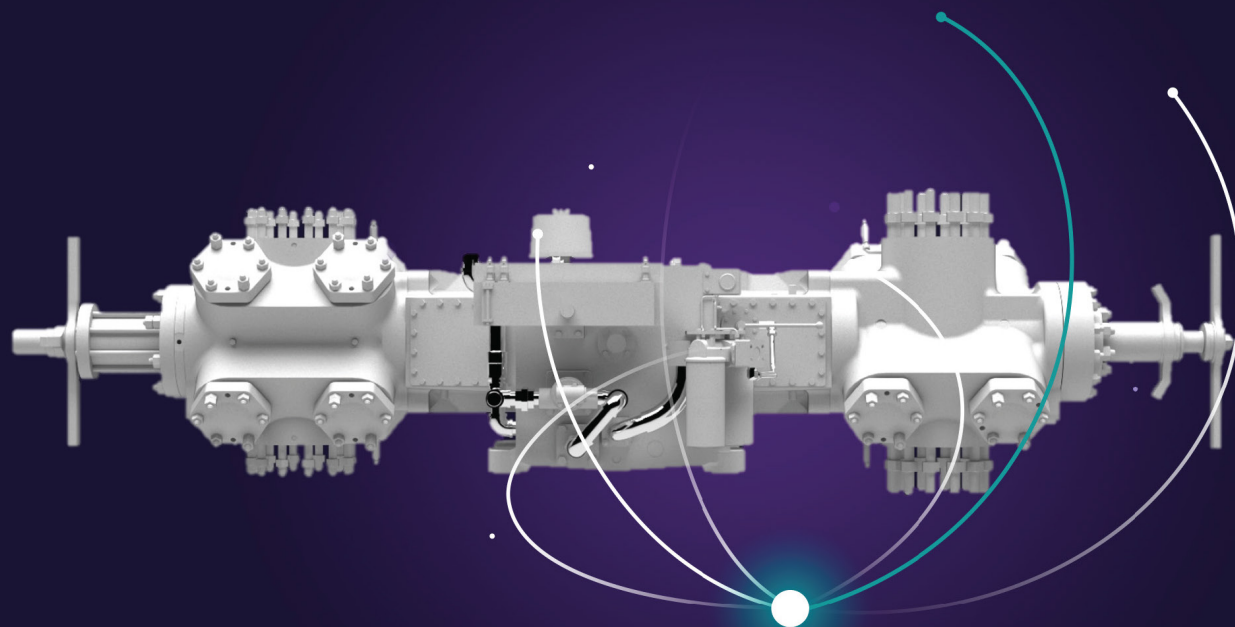


GFP-28

HSRC standard quality assurance plan



Description	Standard testing	Documentation
Cylinder	Material chemical analysis	Available upon request
	Material mechanical test report	Available upon request
Cylinder assembly	Gas passage hydrostatic test performed at 1.5X of maximum allowable working pressure of cylinder	QAF-014
	Coolant passage hydrostatic test performed at 1.5X of maximum allowable working pressure of cylinder (if cylinder is equipped with coolant jacket)	QAF-014
	Gas passage pneumatic leak test performed at maximum allowable working pressure of cylinder (if required for application or client specifications)	QAF-014
	Verification of stud projection in cylinder	QAF-339
	Verification of assembly torque	QAF-339
	Visual inspection of cylinder	QAF-339
Frame	Material chemical analysis	Available upon request
	Material mechanical test report	Available upon request
	Frame sump leak test	Available upon request
Frame assembly	1 Hour no load mechanical run test at operating speed	QAF-262
	Crank pin diametrical clearances	QAF-262
	Crankshaft thrust	QAF-262
	Main bearing top clearance	QAF-262
	Crosshead pin clearances	QAF-262
	Reciprocating assembly weights	QAF-262
	Cylinder bore dimension	QAF-262
	Piston diameter	QAF-262
	Piston combination ring clearances	QAF-262
	Piston end clearances	QAF-262
	Piston rod runout	QAF-262
	Visual inspection of critical surfaces	QAF-262
	Verification of assembly torque	QAF-262
Connecting rod	Material chemical analysis	Available upon request
	Material mechanical test report	Available upon request
Crankshaft	Dimensional inspection	Available upon request
	Material chemical analysis	Available upon request
	Material mechanical test report	Available upon request
	Magnetic particle inspection	Available upon request
	Ultrasonic inspection	Available upon request
Crosshead	Material chemical analysis	Available upon request
	Material mechanical test report	Available upon request
Crosshead pin	Material chemical analysis	Available upon request
	Material mechanical test report	Available upon request
	Magnetic particle inspection	Available upon request
Outer head	Material chemical analysis	Available upon request
	Material mechanical test report	Available upon request
Piston	Material chemical analysis	Available upon request
	Material mechanical test report	Available upon request
Piston rod nut	Material chemical analysis	Available upon request
	Material mechanical test report	Available upon request

NOTE: If quality provisions are desired that are different from the Standard Quality Assurance Plan listed above, please identify them at time of order. Examples of additional quality provisions are witnessed testing, non-standard component test reports, non-standard non-destructive testing, or supplying documentation that is not provided as standard.

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